

# Notes on Anand & Christov Article

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*NB:* Notes for Chapter 3 from Anand & Christov's Revisiting steady viscous flow of a generalized Newtonian fluid through a slender elastic tube using shell theory. Anything in bold purple has a footnote with a question.

## FLUID MECHANICS PROBLEM

Using cylindrical coordinates  $(r, \theta, z)$  with fluid velocity  $\mathbf{v} = (v_r, v_\theta, v_z)$ , we have the following assumptions:

- (i) Steady flow:  $\frac{\partial}{\partial t} = 0$ .
- (ii) Axisymmetric flow without swirl:  $\frac{\partial}{\partial \theta} = 0$  and  $v_\theta = 0$ .
- (iii) Slender tube:  $\ell \gg a \iff \lambda := \frac{a}{\ell} \ll 1$ .

The conservation law in cylindrical coordinates  $(r, \theta, z)$  reads,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

which simplifies under assumption (i) and (ii),

$$\frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

and further **that**<sup>1</sup>,

$$\frac{1}{r} \frac{\partial}{\partial r}(r v_r) + \frac{\partial v_z}{\partial z} = 0. \tag{1}$$

Introducing the non-dimensional variables:

$$\bar{r} = \frac{r}{a}, \quad \bar{z} = \frac{z}{\ell}, \quad \bar{v}_{\bar{r}} = \frac{v_r}{\mathcal{V}_r}, \quad \bar{v}_{\bar{z}} = \frac{v_z}{\mathcal{V}_z}, \quad \bar{p} = \frac{p}{\mathcal{P}_c} \tag{2}$$

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<sup>1</sup>Are we under the assumption that  $\rho$  is constant?

using the chain rule we find that,

$$\frac{\partial}{\partial r} = \frac{\partial}{\partial \bar{r}} \frac{\partial \bar{r}}{\partial r} = \frac{1}{a} \frac{\partial}{\partial \bar{r}}, \quad \frac{\partial}{\partial z} = \frac{\partial}{\partial \bar{z}} \frac{\partial \bar{z}}{\partial z} = \frac{1}{\ell} \frac{\partial}{\partial \bar{z}}$$

which allows us to write (1) in terms of the new variables:

$$\frac{1}{a\bar{r}} \frac{1}{a} \frac{\partial}{\partial \bar{r}} (a\bar{r} \mathcal{V}_r \bar{v}_{\bar{r}}) + \frac{1}{\ell} \frac{\partial}{\partial \bar{z}} (\mathcal{V}_z \bar{v}_{\bar{z}}) = 0 \iff \frac{\mathcal{V}_r}{\bar{r}} \frac{\partial}{\partial \bar{r}} (\bar{r} \bar{v}_{\bar{r}}) + \lambda \mathcal{V}_z \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} = 0 \quad (3)$$

which we can use to find  $\mathcal{V}_r \equiv \lambda \mathcal{V}_z$ , since this will give us the dimensionless,

$$\frac{1}{\bar{r}} \frac{\partial}{\partial \bar{r}} (\bar{r} \bar{v}_{\bar{r}}) + \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} = 0$$

Consequently, as  $\lambda \ll 1$ , we will have that  $\mathcal{V}_z \gg \mathcal{V}_r$ ; this means the  $z$  scale is much larger than the  $r$  scale which gives us that (to first **order**<sup>2</sup> in  $\lambda$  as  $v_z = \mathcal{O}(1)$  and  $v_r = \mathcal{O}(\lambda)$ ),

$$\frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} = 0$$

and thus  $\bar{\mathbf{v}} = \bar{v}_{\bar{z}}(\bar{r}) \hat{\mathbf{k}}$ .

The components of the momentum conservation law, again in cylindrical coordinates, take the following form:

$r$ -component:

$$\rho \left( \frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) = -\frac{\partial p}{\partial r} + \rho g_r + \left( \frac{1}{r} \frac{\partial (r \tau_{rr})}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta r}}{\partial \theta} - \frac{\tau_{\theta\theta}}{r} + \frac{\partial \tau_{zr}}{\partial z} \right)$$

which under assumptions **(i) and (ii)**<sup>3</sup> simplifies to,

$$0 = \frac{1}{r} \frac{\partial}{\partial r} (r \tau_{rr}) + \frac{\partial \tau_{zr}}{\partial z} - \frac{\partial p}{\partial r} \quad (4a)$$

$\theta$ -component:

$$\rho \left( \frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + v_z \frac{\partial v_\theta}{\partial z} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \rho g_\theta + \left( \frac{1}{r^2} \frac{\partial (r^2 \tau_{r\theta})}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{z\theta}}{\partial z} \right)$$

which under assumptions (i) and (ii) in fact vanishes to zero identically.

<sup>2</sup>Are we taking leading order  $\mathcal{O}(\lambda)$  to mean terms containing powers strictly lower than  $\lambda$ ?

<sup>3</sup>Are we assuming laminar flow? If so where did this come from?

$z$ -component:

$$\rho \left( \frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \rho g_z + \left( \frac{1}{r} \frac{\partial(r\tau_{rz})}{\partial r} + \frac{1}{r} \frac{\partial\tau_{\theta z}}{\partial \theta} + \frac{\partial\tau_{zz}}{\partial z} \right)$$

which under assumptions **(i) and (ii)**<sup>4</sup> simplifies to,

$$0 = \frac{1}{r} \frac{\partial}{\partial r}(r\tau_{rz}) + \frac{\partial\tau_{zz}}{\partial z} - \frac{\partial p}{\partial z} \quad (4b)$$

Now computing the components of  $\bar{\bar{\gamma}}$ , the shear strain rate **tensor**<sup>5</sup>:

$$\dot{\gamma}_{rr} = \frac{\partial v_r}{\partial r} = \frac{\lambda \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \quad (5a)$$

$$\dot{\gamma}_{r\theta} = \frac{\partial v_r}{\partial \theta} + \frac{\partial v_\theta}{\partial r} = 0$$

$$\dot{\gamma}_{rz} = \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} = \frac{\lambda^2 \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{z}} + \frac{\mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \quad (5c)$$

$$\dot{\gamma}_{\theta r} = \frac{\partial v_\theta}{\partial r} + \frac{\partial v_r}{\partial \theta} = 0$$

$$\dot{\gamma}_{\theta\theta} = \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_\theta}{\partial \theta} = 0$$

$$\dot{\gamma}_{\theta z} = \frac{\partial v_\theta}{\partial z} + \frac{\partial v_z}{\partial \theta} = 0$$

$$\dot{\gamma}_{zr} = \dot{\gamma}_{rz}$$

$$\dot{\gamma}_{z\theta} = \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} = 0$$

$$\dot{\gamma}_{zz} = \frac{\partial v_z}{\partial z} + \frac{\partial v_z}{\partial z} = 2 \frac{\partial v_z}{\partial z} = 2 \frac{\lambda \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \quad (5b)$$

having used the dimensionless variables from (2).

Observe that,

$$\frac{a}{\mathcal{V}_z} \dot{\gamma}_{zz} = 2\lambda \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} = \mathcal{O}(\lambda), \quad \frac{a}{\mathcal{V}_z} \dot{\gamma}_{rr} = \lambda \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} = \mathcal{O}(\lambda)$$

whist for the other non-zero components,

$$\frac{a}{\mathcal{V}_z} \dot{\gamma}_{zr} = \frac{a}{\mathcal{V}_z} \dot{\gamma}_{rz} = \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} + \lambda^2 \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{z}} = \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} + \mathcal{O}(\lambda^2)$$

Therefore, to leading order in  $\lambda$ , we have two non-zero **components**<sup>6</sup> for the shear strain rate tensor.

<sup>4</sup>Did this come from the upshot of the continuity equation? If not where?

<sup>5</sup>Why is it not  $\dot{\gamma}_{rr} = \frac{\partial v_r}{\partial r} + \frac{\partial v_r}{\partial r} = 2 \frac{\partial v_r}{\partial r} = 2 \frac{\lambda \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}}$

<sup>6</sup>Which two components?

Using the power law model,

$$\bar{\tau} = \eta \bar{\dot{\gamma}} \quad (6)$$

where we have that  $\eta(\bar{\dot{\gamma}})$  is a function of the **invariants of the shear strain rate tensor**, **specifically depending on the second invariant**<sup>7</sup>  $\frac{1}{2}II$ ,

$$\eta(\bar{\dot{\gamma}}) = m \left| \frac{1}{2}II \right|^{(n-1)/2} \quad (7)$$

where  $m$  is the consistency factor and  $n$  is the power law index.

Under assumption (ii), we **have**<sup>8</sup>

$$\begin{aligned} \frac{1}{2}II &= 2 \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 + \left( \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right)^2 \\ &= 2 \left( \frac{\lambda \mathcal{V}_z}{a} \right)^2 \left[ \left( \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \right)^2 + \left( \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \right)^2 \right] + \left( \frac{\mathcal{V}_z}{a} \right)^2 \left( \lambda^2 \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{z}} + \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right)^2 \\ &= \left( \frac{\mathcal{V}_z}{a} \right)^2 \left( \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right)^2 + \mathcal{O}(\lambda^2) \end{aligned} \quad (8)$$

Now using the power law in components,

$$\tau_{ij} = \eta(\bar{\dot{\gamma}}) \left( \frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$$

we can find,

$$\tau_{rz} = \tau_{zr} = \eta(\bar{\dot{\gamma}}) \left( \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) \quad (9)$$

Introducing  $\mathcal{T}_c$  as the scale for shear stress:

$$\begin{aligned} \tau_{rz} = \mathcal{T}_c \bar{\tau}_{\bar{r}\bar{z}} = \tau_{zr} = \mathcal{T}_c \bar{\tau}_{\bar{z}\bar{r}} &= m \left| \frac{1}{2}II \right|^{(n-1)/2} \left( \frac{\lambda^2 \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{z}} + \frac{\mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right) \\ &= m \left( \frac{\mathcal{V}_z}{a} \right)^n \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} + \mathcal{O}(\lambda^{n-1}) \end{aligned} \quad (10)$$

which suggests a choice of scale as,

$$\mathcal{T}_c = m \left( \frac{\mathcal{V}_z}{a} \right)^n \quad (11)$$

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<sup>7</sup>Excuse me?

<sup>8</sup>Shouldn't the first term in the second line be  $\left( \frac{\lambda \mathcal{V}_z}{a} \right)^2 \left[ 2 \left( \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \right)^2 + \left( \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \right)^2 \right]$ ?

Applying the same reasoning to the other components of the shear stress **gives**<sup>9</sup>,

$$\bar{\tau}_{\bar{z}\bar{z}} = \lambda \mathcal{T}_c \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \quad (12a)$$

$$\bar{\tau}_{\bar{r}\bar{r}} = \lambda \mathcal{T}_c \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \quad (12b)$$

which are  $\mathcal{O}(\lambda)$  and are negligible compared to the components in (10).

Next, **nondimensionalising**<sup>10</sup> (4b):

$$0 = \mathcal{T}_c \frac{1}{\bar{r}} \frac{\partial}{\partial \bar{r}} (\bar{r} \bar{\tau}_{\bar{r}\bar{z}}) + \lambda^2 \mathcal{T}_c \frac{\partial \bar{\tau}_{\bar{z}\bar{z}}}{\partial \bar{z}} - \lambda \mathcal{P}_c \frac{\partial \bar{p}}{\partial \bar{z}} \quad (13)$$

which **requires**<sup>11</sup>,

$$\mathcal{P}_c = \frac{\mathcal{T}_c}{\lambda}$$

or in other words  $\mathcal{P}_c \gg \mathcal{T}_c$ , and the gradient of the normal stress is negligible to leading order in  $\lambda$ .

**Nondimensionalising**<sup>12</sup> (4a),

$$0 = \lambda^2 \mathcal{T}_c \frac{1}{\bar{r}} \frac{\partial}{\partial \bar{r}} (\bar{r} \bar{\tau}_{\bar{r}\bar{z}}) + \lambda^2 \mathcal{T}_c \frac{\partial \bar{\tau}_{\bar{z}\bar{z}}}{\partial \bar{z}} - \mathcal{T}_c \frac{\partial \bar{p}}{\partial \bar{z}} \quad (15)$$

and thus to leading order in  $\lambda$ ,

$$0 = \frac{\partial \bar{p}}{\partial \bar{r}} \quad (16)$$

and we are left with  $\bar{p} = \bar{p}(\bar{z})$  and also,

$$\mathcal{P}_c = \frac{m \ell \mathcal{V}_z^n}{a^{n+1}} \quad (17)$$

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<sup>9</sup>Why is it not:

$$\begin{aligned} \tau_{zz} &= \eta(\bar{\gamma}) \left( \frac{\partial v_z}{\partial z} + \frac{\partial v_z}{\partial z} \right) = 2\eta(\bar{\gamma}) \frac{\partial v_z}{\partial z}, \quad \tau_{rr} = \eta(\bar{\gamma}) \left( \frac{\partial v_r}{\partial r} + \frac{\partial v_r}{\partial r} \right) = 2\eta(\bar{\gamma}) \frac{\partial v_r}{\partial r} \\ \tau_{\bar{z}\bar{z}} &= \mathcal{T}_c \bar{\tau}_{\bar{z}\bar{z}} = m \left| \frac{1}{2} II \right|^{(n-1)/2} \left( 2 \frac{\lambda \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \right) = 2\lambda m \left( \frac{\mathcal{V}_z}{a} \right)^n \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}} \\ \tau_{\bar{r}\bar{r}} &= \mathcal{T}_c \bar{\tau}_{\bar{r}\bar{r}} = m \left| \frac{1}{2} II \right|^{(n-1)/2} \left( 2 \frac{\lambda \mathcal{V}_z}{a} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \right) = 2\lambda m \left( \frac{\mathcal{V}_z}{a} \right)^n \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \\ \bar{\tau}_{\bar{z}\bar{z}} &= 2\lambda \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{z}}, \quad \bar{\tau}_{\bar{r}\bar{r}} = 2\lambda \left| \frac{\partial \bar{v}_{\bar{z}}}{\partial \bar{r}} \right|^{n-1} \frac{\partial \bar{v}_{\bar{r}}}{\partial \bar{r}} \end{aligned}$$

<sup>10</sup>Where did the  $\lambda^2$  come from? Multiplying by  $a$ :

$$0 = \frac{1}{a\bar{r}} \frac{1}{a} \frac{\partial}{\partial \bar{r}} (a\bar{r} \mathcal{T}_c \bar{\tau}_{\bar{r}\bar{z}}) + \frac{\lambda}{a} \frac{\partial (\mathcal{T}_c \bar{\tau}_{\bar{z}\bar{z}})}{\partial \bar{z}} - \frac{\lambda}{a} \frac{\partial (\mathcal{P}_c \bar{p})}{\partial \bar{z}} \iff 0 = \mathcal{T}_c \frac{1}{\bar{r}} \frac{\partial}{\partial \bar{r}} (\bar{r} \bar{\tau}_{\bar{r}\bar{z}}) + \lambda \mathcal{T}_c \frac{\partial \bar{\tau}_{\bar{z}\bar{z}}}{\partial \bar{z}} - \lambda \mathcal{P}_c \frac{\partial \bar{p}}{\partial \bar{z}}$$

<sup>11</sup>Can you explain this in more detail, I understand the general idea but precisely what is going on for these dimensional relations?

<sup>12</sup>Again, why is there an extra  $\lambda$  in the first term?

In a flow-rate-controlled experiment simulation, we can choose a velocity **scale**<sup>13</sup>  $\mathcal{V}_z = \frac{q}{\pi a^2}$  and thus  $\mathcal{P}_c = \frac{m\ell q^n}{a^{3n+1}\pi^n}$ .

For axisymmetric flow, we expect that the axial velocity will attain its maximum along the centreline ( $r = 0$ ), decreasing with the radius until it reaches zero at the tube wall (due to no slip) in this steady flow. This means that  $|\frac{\partial \bar{v}_z}{\partial \bar{r}}| = -\frac{\partial \bar{v}_z}{\partial \bar{r}}$ , and (10) reads,

$$\bar{\tau}_{\bar{r}\bar{z}} = - \left( -\frac{\partial \bar{v}_z}{\partial \bar{r}} \right)^n \quad (18)$$

Ignoring terms  $\mathcal{O}(\lambda^2)$ , (13) reads,

$$\frac{1}{\bar{r}} \frac{\partial}{\partial \bar{r}} \left[ \bar{r} \left( -\frac{\partial \bar{v}_z}{\partial \bar{r}} \right)^n \right] = -\frac{d\bar{p}}{d\bar{z}} \quad (19)$$

which can be solved with the no slip and regularity conditions to yield,

$$\bar{v}_z = \left( -\frac{1}{2} \frac{d\bar{p}}{d\bar{z}} \right)^{1/n} \frac{\bar{R}^{1+1/n} - \bar{r}^{1+1/n}}{1 + 1/n}$$

where  $\bar{R} = R/a$ .

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<sup>13</sup>Where is this coming from?